DATA JOB MARKET
SKILL DEMAND ANALYSIS

Stage 4 – Documentation, Insights and Presentation

Job Market Analytics | Python + Web Scraping + Power BI

Project Domain	Job Market Analytics
Primary Dataset	Datanemics – Data Job Postings
Additional Source	MyJobMag web-scraped job postings
Final Dataset	3,518 records and 11 features

1. Project Overview

The objective of this project is to analyze data job postings and identify meaningful patterns in job roles, skills, seniority levels, locations, work arrangements and salary. The analysis is intended to help job seekers understand the skills and qualifications currently valued in the data job market.

Objectives

- Identify the most in-demand skills mentioned in data-related job postings.
- Analyze the distribution of different data-related job roles.
- Examine job demand across cities and countries.
- Analyze job postings by seniority level.
- Understand the distribution of remote, hybrid and onsite opportunities.
- Identify relationships between skills, job roles, experience levels and salary.
- Generate actionable insights for skill development and career planning.

Dataset and Methodology

The project uses the Datanemics Data Job Postings dataset containing 3,500 records and 11 features. An additional 18 job postings were collected from MyJobMag using web scraping, resulting in a combined dataset of 3,518 records and 11 aligned features.

Stage	Activity	Output
Stage 1	Problem definition and dataset selection	Project objective and source data
Stage 2	Cleaning and preprocessing	Cleaned and standardized dataset
Stage 3	EDA and visualization	Univariate, bivariate and multivariate analysis
Stage 4	Dashboard, insights and presentation	Power BI report and final project story

2. Power BI Dashboard

The cleaned and combined dataset was exported from Python as a CSV file and used as the data source for Power BI. The dashboard brings the major findings from the exploratory analysis into one interactive report.

Dashboard Components Implemented

The overview page combines KPI cards, Top 10 Job Roles and Skills, seniority and job-type distributions, average minimum salary by seniority, and a Country-to-City drill-down. Country, Seniority and Job Type dropdown slicers are synchronized across pages. The Job Role Details page contains a posting table, skills, seniority, work-type and country breakdowns, Title-based drill-through, and a Reset to All Data bookmark.

Dashboard Page 1 - Job Market Overview

This overview page summarizes the job-posting dataset using KPI cards, skills and role rankings, seniority and work-type distributions, salary by seniority, and a Country-to-City drill-down chart.



Figure 1. Power BI Page 1 - Job Market Overview (updated screenshot supplied by the project owner).

Dashboard Page 2 - Job Role Details

The detail page supports drill-through by job title, a reset-to-all-data bookmark, a posting details table, and additional role-level breakdowns. The screenshot shows the page in its current reset/all-data state.

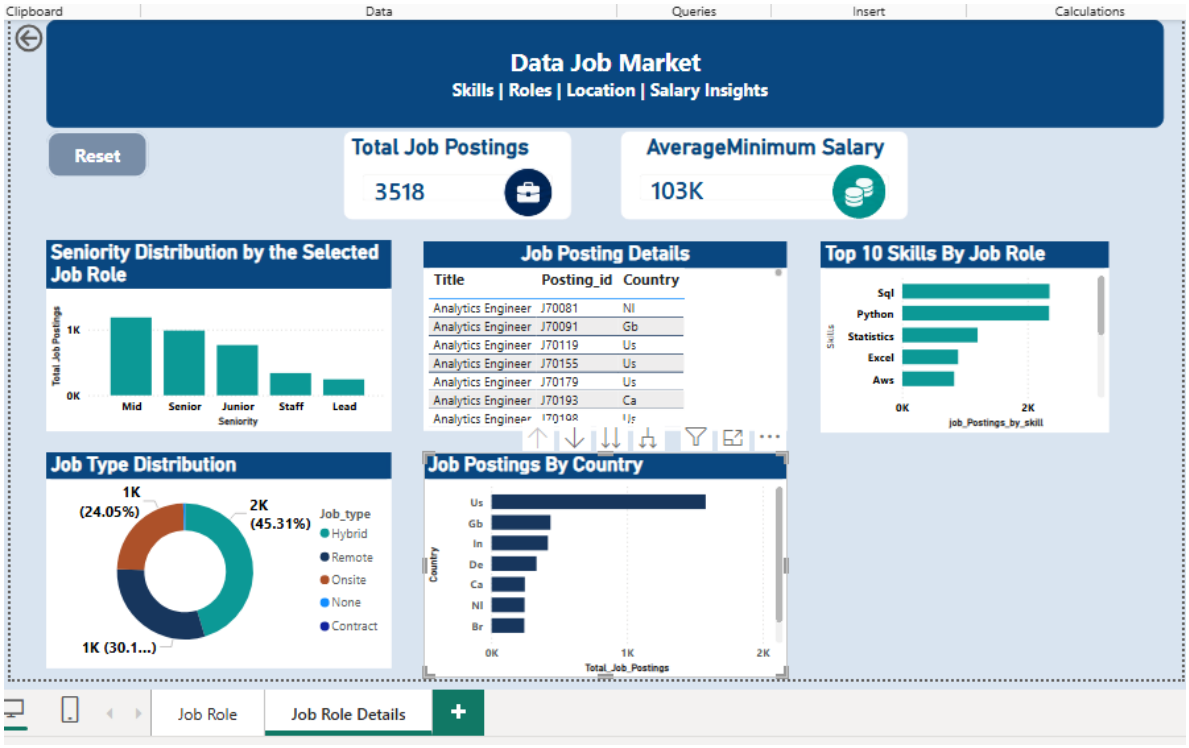


Figure 2. Power BI Page 2 - Job Role Details (updated screenshot supplied by the project owner).

3. Key Findings and Insights from Power BI and Python EDA

3.1 Dashboard Snapshot

The current Power BI overview shows 3,518 total job postings, 51 unique job titles and 8 country categories. The Average Minimum Salary card displays approximately 103K; add an explicit USD label and validate the salary assumptions before interpreting this as a precise market-wide benchmark.

3.2 Most In-Demand Skills

SQL and Python are the strongest visible skill signals in the Power BI chart. Statistics and Excel follow, while AWS, Airflow and PyTorch are also visible. The earlier Python top-skills analysis additionally identified Looker, Tableau and Terraform among the leading skills. The skills chart is scrollable in the current layout, so not every top-10 category is visible at once.

3.3 Job Role Demand

Data Analyst is the leading role in the Top 10 Job Roles visual. Senior Data Analyst and Data Engineer are also prominent, with Junior Data Analyst and Data Scientist appearing in the visible leading group. The Python role-by-seniority heatmap showed Data Analyst and Data Engineer postings concentrated in Mid-level roles, while Senior Data Analyst and Senior Data Engineer were more strongly represented at Senior level.

3.4 Seniority Distribution

Mid-level postings are the largest seniority group in the current dashboard, followed by Senior and Junior. Staff and Lead postings are less frequent. Python analysis also showed that job-role demand varies by seniority: Data Analyst and Data Engineer had strong Mid-level counts, while explicitly senior-titled roles were more represented in the Senior group.

3.5 Salary and Seniority

The Average Minimum Salary by Seniority chart shows a general increase from Junior through Mid and Senior to Staff and Lead, with Lead and Staff among the highest averages and Junior the lowest. The Python salary box plot also identified potential high-salary outliers across seniority groups. These should be investigated rather than automatically removed, as some may be legitimate senior or specialized positions.

3.6 Work Arrangement

The current donut chart shows Hybrid as the largest displayed category (45.31%), followed by Remote (30.19%) and Onsite (24.05%). The legend also includes None and Contract. Treat these categories carefully: None means unspecified or missing, while Contract describes an employment type rather than a work arrangement. Standardize the categories before presenting the chart as a definitive split of remote, hybrid and onsite work.

3.7 Skills Across Seniority

The Python multivariate skill chart showed SQL and Python with strong demand across Junior, Mid, Senior, Staff and Lead postings, with Mid-level counts highest for most skills. Statistics was also prominent, and other technical and BI skills appeared across multiple career levels. This supports a learning plan built around SQL, Python, statistics and Excel, followed by tools relevant to a chosen role.

3.8 Location Pattern and Drill-Down

The current country chart shows the US category as the largest source of postings, followed by smaller country categories. Some labels appear as abbreviations, so use standardized full country names wherever possible. The Country → City hierarchy is intended to let viewers drill from a country summary into its cities; verify the city level before interpreting or presenting a screenshot of the drill-down.

3.9 Dashboard Interactivity

The report contains three dropdown slicers (Country, Seniority and Job Type), synchronized between the two pages. Page 1 supports Country-to-City drill-down. Page 2 uses Title as the drill-through field and has a Reset to All Data bookmark/button so viewers can return to the full-data view. Keep all filters is disabled for drill-through to avoid unintentionally carrying unrelated source-page filters into the detail page.

3.10 Data Quality and Interpretation Notes

The Page 1 salary card currently displays approximately 103K without an explicit USD label, and the Average Minimum Salary card on Page 2 appears blank in the supplied screenshot; verify its measure and filter context before submission. The source workflow imputed some missing salary values in scraped MyJobMag records using a median from the Datanemics data, and converted some Nigerian monthly salaries to annual USD using an assumed exchange rate. As a result, the blended average salary is indicative and should not be presented as a precise cross-country market benchmark. The Job Type field also currently mixes work arrangements (Remote, Hybrid, Onsite), missing/unspecified values (None) and Contract; standardize or separate employment type from work arrangement before finalizing categorical percentages. Some location labels are abbreviated and should be cleaned for presentation.

4. Recommendations

1. Prioritize SQL and Python. They are the strongest visible skill signals and should form the core of a data analyst learning plan.
2. Strengthen analytical fundamentals. Learn statistics and Excel alongside programming and database skills.
3. Build skills progressively by career level. Combine core analytics skills with tools such as AWS, Airflow, PyTorch or BI platforms when they fit the target role.
4. Prepare for a specific role. Compare the requirements of Data Analyst, Data Engineer, Data Scientist and related roles before choosing a specialization.
5. Compare work arrangement and location together with role, seniority and salary. Standardize the mixed Job Type categories, validate currency conversions and imputed values, and investigate salary outliers instead of automatically deleting them.

5. Final Story of the Dashboard

The dashboard tells a clear story about the demand for data professionals. The job market contains a wide range of data-related roles, with Data Analyst among the most represented. The strongest skill signals are SQL and Python, supported by analytical skills such as Statistics and Excel and a range of cloud, workflow, machine-learning and business-intelligence technologies.

The seniority analysis indicates that Mid-level opportunities form a major portion of the market. At the same time, salary generally rises with seniority, showing the value of experience and progression. The multivariate analysis demonstrates that skill demand changes across career levels, making it useful to plan skill development according to the target role and seniority.

Overall, the dashboard converts raw job-posting data into practical career information. A job seeker can use it to identify high-demand skills, understand the roles available at different career levels, compare work arrangements and locations, and make more informed decisions about learning and job applications.

6. Conclusion

This project demonstrates an end-to-end analytics workflow: problem definition, dataset selection, web scraping, data cleaning and transformation, Python EDA, data modeling in Power BI, interactive visualization and documentation. The attached screenshots record the current two-page Power BI report and its drill-down, drill-through, synchronized slicers and reset-bookmark behavior.

End of Stage 4 Documentation